The goal of using raw, unfiltered data to improve robustness of the analysis in the context of the *S.T.A.R. Program, the following appropriate VizieR catalogues align with the focus on galaxy properties, star formation rates, and BSD-inspired features.

The program works with galaxy-wide measurements (e.g., log_SFR_Ha, log_Mass_gas, Re_kpc, OH_O3N2_cen) and aims to model star formation rates while incorporating cosmological influences through features like cosmo_rank and L_cosmo_s*.

Using raw data to preserve observational variability and reduce biases from preprocessing, prioritizing catalogues with minimally processed measurements (e.g., fluxes, magnitudes) relevant to galaxies.

With direct links to these catalogues on VizieR, ensuring they can be accessed for analysis.

The program is designed to:
Model log_SFR_Ha (star formation rate derived from Hα emission) for galaxies.

Use features like log_Mass_gas, Re_kpc (effective radius), and metallicity indicators (OH_O3N2_cen).

Incorporate BSD features (cosmo_rank, L_cosmo_s*) that depend on galaxy properties like mass and size.

Improve robustness by using raw data, meaning to preserve observational noise (e.g., measurement errors, systematic effects) rather than relying on preprocessed or derived quantities.

Raw Data in Astronomy: In this context, "raw unfiltered data" refers to measurements closer to direct observations—such as fluxes, magnitudes, or spectral line intensities—before they are heavily processed into derived quantities like SFRs or metallicities.

While no catalogue provides truly raw telescope data (e.g., photon counts), some VizieR catalogues offer minimally processed data (calibrated but not aggregated or cleaned), which aligns with the goal.

Requirements for Catalogues:

Object Type: Must focus on galaxies, not stellar systems like cataclysmic variables (e.g., III/286 was unsuitable).

Data Type: Should provide raw or minimally processed measurements (e.g., emission line fluxes, photometry) relevant to star formation, mass, or metallicity.

Cross-Matching: Must include coordinates (RA, Dec) to merge with your dataset (objra_y, objdec).

Robustness: Should preserve observational variability (e.g., include errors, avoid heavy outlier removal).

VizieR catalogues that meet the criteria, with explanations of their relevance, the type of raw data they provide, and direct links to access them.

Focusing on SDSS-based catalogues (and complementary surveys like 2MASS and GAMA) which provide raw measurements for galaxies.

1. SDSS DR16 Spectroscopic Data (V/154)

Why It's Suitable:

Contains spectroscopic measurements for galaxies, including raw emission line fluxes (e.g., Hα, Hβ, [OIII], [NII]), which are directly relevant to your target (log_SFR_Ha) and features (e.g., OH_O3N2_cen).

Provides minimally processed data: fluxes are calibrated (e.g., corrected for instrumental effects) but not aggregated or heavily cleaned, preserving observational noise.

Includes measurement errors (e.g., e_flux_6563), which can enhance robustness by exposing the model to uncertainty.

Matches the SDSS origin dataset, increasing the likelihood of successful cross-matching.

Key Columns:

RA, Dec (for cross-matching with objra_y, objdec).

specz (spectroscopic redshift, useful for distance calculations).

flux_6563 (Hα flux, can be used to recompute log_SFR_Ha).

flux_4861 (Hβ flux, for extinction correction via Hα/Hβ ratio).

flux_5007 ([OIII] 5007 flux, for metallicity via O3N2).

flux_6584 ([NII] 6584 flux, for metallicity).

e_flux_6563 (error in Hα flux, captures observational uncertainty).

How It Improves Robustness:

Raw fluxes include measurement errors and systematic effects (e.g., extinction, aperture corrections), making the model more resilient to real-world data imperfections.

Recomputing log_SFR_Ha and metallicity from fluxes introduces variability not present in preprocessed values.

Link to Catalogue:

Access V/154 on VizieR: http://vizier.cds.unistra.fr/viz-bin/VizieR?-source=V/154

Query by coordinates to match the dataset or download the full table (full table chosen)

2. SDSS DR12 Photometric Catalogue (V/147)
Why It's Suitable:

Provides photometric measurements (magnitudes in u, g, r, i, z bands) for millions of galaxies from SDSS DR12.

Raw Data: Offers PSF, Petrosian, and model magnitudes directly from observations, along with errors, without heavy processing beyond calibration.

Complements spectroscopic data by adding optical photometry, which can be used to derive galaxy colors (e.g., g-r, u-g) as proxies for stellar population age and star formation history.

Key Columns:

RA, Dec.

petroMag_u, petroMag_g, petroMag_r, petroMag_i, petroMag_z (Petrosian magnitudes in SDSS bands).

petroMagErr_u, petroMagErr_g, etc. (magnitude errors).

z (photometric redshift, if spectroscopic redshift is unavailable).

How It Improves Robustness:

Raw magnitudes introduce variability due to dust, observational conditions, and measurement errors, reducing overfitting to cleaned data.

Galaxy colors can capture additional variance in star formation activity, complementing log_SFR_Ha.

Link to Catalogue:

Access V/147 on VizieR: http://vizier.cds.unistra.fr/viz-bin/VizieR?-source=V/147

Query by position or download subsets (PhotoObjAll table chosen)

3. 2MASS Extended Source Catalogue (II/246)

Why It's Suitable:

Provides near-infrared photometry (J, H, Ks bands) for extended sources (mostly galaxies) from the Two Micron All Sky Survey (2MASS).

Raw Data: Includes magnitudes and fluxes with errors, minimally processed beyond calibration.

Near-infrared data is less affected by dust extinction, providing a complementary view of stellar mass (correlates with log_Mass in the dataset).

Key Columns:
RA, Dec.

Jmag, Hmag, Kmag (extrapolated magnitudes in J, H, Ks bands).

e_Jmag, e_Hmag, e_Kmag (magnitude errors).

r_ext (extrapolated radius, a proxy for galaxy size).

How It Improves Robustness:

Adds a different wavelength regime (near-infrared), exposing the model to galaxies with varying dust content.

Magnitude errors capture observational uncertainty, enhancing generalizability.

Link to Catalogue:

Access II/246 on VizieR: http://vizier.cds.unistra.fr/viz-bin/VizieR?-source=II/246

Focusing on the main table (xsc) for extended sources.

4. GAMA DR3 Spectroscopic and Photometric Data (II/356)

Why It's Suitable:

The Galaxy And Mass Assembly (GAMA) survey provides both spectroscopic and multi-wavelength photometric data for galaxies, overlapping with SDSS but extending to deeper magnitudes.

Raw Data: Includes emission line fluxes (e.g., Hα, [NII]) and photometry across 20 bands (UV to far-IR), minimally processed.

Particularly useful for recomputing log_SFR_Ha and adding multi-wavelength features.

Key Columns:

RA, Dec.

Flux_Ha (Hα flux from spectroscopy).

Flux_NII_6584 ([NII] 6584 flux).

u_mag, g_mag, r_mag, etc. (photometry from SDSS and other surveys).

e_Flux_Ha (error in Hα flux).

How It Improves Robustness:

Combines spectroscopy and photometry, providing a comprehensive view of galaxy properties with observational noise intact.

Multi-wavelength data helps capture diverse physical processes (e.g., star formation, dust), improving model generalizability.

Link to Catalogue:

Access II/356 on VizieR: http://vizier.cds.unistra.fr/viz-bin/VizieR?-source=II/356

GamaSpec (spectroscopy) and GamaPhoto (photometry)